EC015
Chemistry I
Semester I
Session 2026/2027
2 hours

EC015
Kimia I
Semester I
Sesi 2026/2027
2 jam

KOLEJ MATRIKULASI KEJURUTERAAN KEDAH
KECEMERLANGAN MENJANA TEKNOKRAT KOMPETEN

PROGRAM JELAJAH PRESTASI KMKK

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
PRE-MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

SOALAN	MARKAH
1	
2	
3	
4	
5	
6	
JUMLAH	

NAMA PELAJAR	
TUTORIAL	
NO. MATRIK	
NAMA PENSYARAH	

Kertas soalan ini mengandungi 10 halaman bercetak.
This question paper consists of 10 printed pages.

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INSTRUCTIONS TO CANDIDATE:

This question paper consists of **6** questions. Answer **all** questions.
Answers should be written in the answer booklet provided.
All steps must be shown clearly. Use a new page for each question.
Maximum mark awarded is shown at the end of each question or section.
The use of non-programmable scientific calculator is permitted.

***ARAHAN KEPADA CALON*:**

*Kertas soalan ini mengandungi **6** soalan. Jawab **semua** soalan.*
Jawapan hendaklah ditulis pada buku jawapan yang disediakan.
Semua langkah kerja hendaklah ditunjukkan dengan jelas. Gunakan muka surat baharu bagi nombor soalan yang berbeza.
Markah maksimum yang diperuntukkan ditunjukkan pada hujung setiap soalan atau bahagian soalan.
Penggunaan kalkulator saintifik yang tidak boleh diprogramkan dibenarkan.

TABLE OF RELATIVE ATOMIC MASSES

Element	Symbol	Proton Number	Relative Atomic Mass
Aluminium	Al	13	27.0
Silver	Ag	47	107.9
Argon	Ar	18	40.0
Arsenic	As	33	74.9
Gold	Au	79	197.0
Barium	Ba	56	137.3
Beryllium	Be	4	9.0
Bismuth	Bi	83	209.0
Boron	B	5	10.8
Bromine	Br	35	79.9
Iron	Fe	26	55.9
Flourine	F	9	19.0
Phosphorus	P	15	31.0
Helium	He	2	4.0
Mercury	Hg	80	200.6
Hydrogen	H	1	1.0
Iodine	I	53	126.9
Cadmium	Cd	48	112.4
Potassium	K	19	39.1
Calcium	Ca	20	40.1
Carbon	C	6	12.0
Chlorine	Cl	17	35.5
Cobalt	Co	27	58.9
Krypton	Kr	36	83.8
Chromium	Cr	24	52.0
Copper	Cu	29	63.6
Lithium	Li	3	6.9
Magnesium	Mg	12	24.3
Manganese	Mn	25	54.9
Sodium	Na	11	23.0
Neon	Ne	10	20.2
Nickel	Ni	28	58.7
Nitrogen	N	7	14.0
Oxygen	O	8	16.0
Platinum	Pt	78	195.1
Lead	Pb	82	207.2
Protactinium	Pa	91	231.0
Radium	Ra	88	226.0
Radon	Rn	86	222.0
Rubidium	Rb	37	85.5
Selenium	Se	34	79.0
Cerium	Ce	58	140.1
Cesium	Cs	55	132.9
Silicon	Si	14	28.1
Scandium	Sc	21	45.0
Tin	Sn	50	118.7
Antimony	Sb	51	121.8
Strontium	Sr	38	87.6
Sulfur	S	16	32.1
Uranium	U	92	238.0
Tungsten	W	74	183.9
Zinc	Zn	30	65.4

LIST OF SELECTED CONSTANT VALUES

Ionisation constant for water at 25 ^{0}C	K_w	=	1.00 x 10^{-14} mol^2 dm $^{-6}$
Molar volume of gases	V_m	=	22.4 dm^3 mol^{-1} at STP
		=	24 dm^3 mol^{-1} at room temperature
Speed of light in a vacuum	c	=	3.0 x 10^8 m s^{-1}
Specific heat of water		=	4.18 kJ kg^{-1} K^{-1}
		=	4.18 J g^{-1} K^{-1}
		=	4.18 J g^{-1} ^{0}C^{-1}
Avogadro's number	N_A	=	6.02 x 10^{23} mol^{-1}
Faraday constant	F	=	9.65 x 10^4 C mol^{-1}
Planck constant	h	=	6.6256 x 10^{-34} J s
Rydberg constant	R_H	=	1.097 x 10^7 m^{-1}
		=	2.18 x 10^{-18} J
Molar of gases constant	R	=	8.314 J K^{-1} mol^{-1}
		=	0.08206 L atm mol^{-1} K^{-1}
Density of water	ρ	=	1 g cm^{-3}
Freezing point of water		=	0.00 ^{0}C
Vapour pressure of water	P_{water}	=	23.8 torr

UNIT AND CONVERSION FACTOR

Volume	1 liter = 1 dm^3
	1 mL = 1 cm^3
Energy	1 J = 1 kg m^2 s^{-2} = 1 N m = 10^7 erg
	1 calorie = 4.184 Joule
	1 eV = 1.602 x 10^{-19} J
Pressure	1 atm = 760 mm Hg = 760 torr = 101.325 kPa = 101325 N m^{-2}
Others	1 Faraday (F) = 96500 Coulomb
	1 Newton (N) = 1 kg m s^{-2}

Answer **ALL** questions.

1. (a) Gallium occurs naturally as two isotopes, ^{69}Ga and ^{71}Ga, with relative isotopic masses of 68.926 amu and 70.925 amu respectively. A mass spectrum shows that the relative abundance of ^{69}Ga is 60.11% and ^{71}Ga is 39.89%.

(i) Calculate the relative atomic mass of gallium.

(ii) Write the complete isotopic notation for the cation formed when a ^{69}Ga atom loses three electrons.

[4 *marks*]

(b) A chemist needs to prepare 250.0 mL of a 0.705 M dilute nitric acid solution for an experiment. Available in the laboratory is a stock bottle of concentrated HNO_3 labeled 68.0% by mass with a density of 1.4 gcm^{-3}.

(i) Determine the molality of the concentrated HNO_3 stock solution.

(ii) Calculate the volume (mL) of the concentrated HNO_3 stock solution required to prepare the desired diluted solution.

[6 *marks*]

(c) Balance the following redox equation in basic medium using the ion-electron method:

$$MnO_4^-(aq) + SO_3^{2-}(aq) \rightarrow MnO_2(s) + SO_4^{2-}(aq)$$

[4 *marks*]

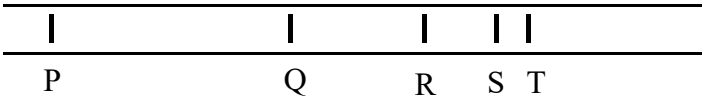
(d) Phosphorus trichloride, PCl_3, is an important chemical intermediate used in the production of pesticides and flame retardants. It is prepared by reacting elemental phosphorus with chlorine gas according to the equation:

$$P_4(s) + 6Cl_2(g) \rightarrow 4PCl_3(l)$$

In a reaction vessel, 15.5g of P_4 is mixed with 45.0g of Cl_2. Calculate the mass (in grams) of the excess reactant remaining after the reaction.

[7 *marks*]

2. (a) The figure below shows the line spectrum of hydrogen atom in the Brackett series.



(i) Calculate the wavelength (in nm) and frequency of the radiation emitted in the formation of line R .

(ii) Which line corresponds to the transition with the highest energy.

[5 *marks*]

(b) Element Z has proton number 29.

(i) Write the electronic configuration of Z^{2+} ion.

(ii) Write a set of quantum number for the seventh electron in highest energy orbital for Z^{2+} ion.

(iii) Explain the anomalous electronic configuration of element Z.

[5 *marks*]

3. (a) NO_2^+ is a key component in the nitration process, essential for the synthesis of explosives, dyes and pharmaceuticals.

(i) Draw two possible Lewis structure for NO_2^+.

(ii) Determine the most plausible Lewis structure in (i). Give your reason.

[6 *marks*]

(b) (i) State the type of hybridisation and show the hybrid process for central atom in ICl_2^-.

(ii) Draw and label the orbital overlapping involved the sigma (σ) bond in ICl_2^-

[8 *marks*]

(c) Methanol, CH_3OH, and dimethyl ether, CH_3OCH_3, have different boiling points.
Explain why methanol has a higher boiling point than dimethyl ether in terms of intermolecular forces.

[3 *marks*]

4. (a) A 6.00 dm^3 rigid vessel contains a mixture of 5.04 g of nitrogen gas, N_2, and 0.640 g of hydrogen gas, H_2, at 27.0 °C. Assume that both gases behave ideally and do not react with each other.

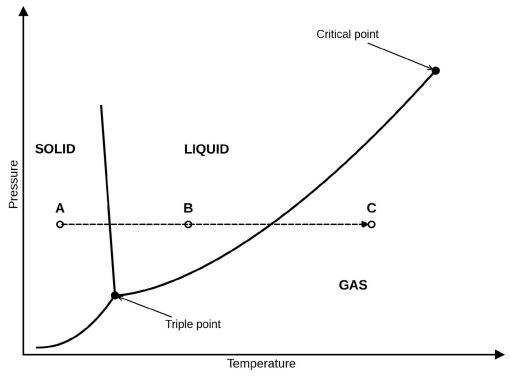
(i) Calculate the total pressure, in Pa, exerted by the gas mixture.

[3 *marks*]

(ii) The vessel is then connected to an evacuated 4.00 dm^3 flask by opening a valve, and the whole system is heated to 87.0 °C. Calculate the partial pressure, in Pa, of H_2 in the new system.

[2 *marks*]

(b) Figure 1 shows a simplified phase diagram of water. Point A is in the solid region, point B is in the liquid region and point C is in the gas region. The line A → B → C represents heating at constant pressure.



(i) Explain why the solid–liquid equilibrium line in the phase diagram of water has a negative slope.

[2 *marks*]

(ii) A student claims that if the heating from A to C is repeated at a pressure below the triple point pressure, the ice will still melt before it turns into gas. Using the phase diagram, justify whether the student's claim is correct.

[2 *marks*]

5. Solid ammonium hydrogen sulfide decomposes according to the equation:

$$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$$

When excess solid NH_4HS was placed in an evacuated flask at 298 K, the total pressure at equilibrium was 0.600 atm.

(a) (i) Calculate the value of Kp and Kc at 298 K.

(ii) In a separate experiment at 298 K, excess $NH_4HS(s)$ was placed in a flask that already contained NH_3 gas at a pressure of 0.500 atm. Calculate the partial pressure of H_2S at equilibrium.

[7 *marks*]

(b) At 318 K, the value of Kp is 0.30 atm^2. By comparing this value with your answer in **(a)(i)**, deduce whether the forward reaction is exothermic or endothermic. Explain your answer.

[2 *marks*]

6. (a) In organic chemistry, a buffer solution plays a critical role when diazonium salts undergo azo coupling reactions. A buffer solution is prepared when 200 mL of 0.15 M nitrous acid, HNO_2 is mixed with 300 mL of 0.20 M sodium nitrite, $NaNO_2$ solution.
[Given the acid dissociation constant, K_a for HNO_2 is 4.95×10^{-4}]

(i) Write the chemical equations when a small amount of acid is added to the above buffer solution.

(ii) Calculate the pH of the prepared buffer solution.

[7 *marks*]

(b) Calcium fluoride, CaF_2 is commonly used in various dental preparations and toothpastes to help strengthen tooth enamel and prevent cavities. The solubility product of CaF_2 is 3.90×10^{-11} at 25°C.

(i) Calculate the molar solubility of CaF_2 in water at 25°C.

(ii) Explain the effect of solubility of CaF_2 if calcium nitrate, $Ca(NO_3)_2$ solution is added to CaF_2.

[7 *marks*]

END OF QUESTION PAPER